APPLICATION OF REINFORCEMENT LEARNING OPTIMISATION METHODOLOGY TO BWR ASSEMBLIES

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ABSTRACT

We perform Artificial Intelligence (AI) driven 3D optimisation of GE14-10x10 BWR assembly using deep reinforcement learning (RL). Radial (2D) fuel optimisation is done using CASMO4 for different types of zones (partial-length, power shaping) to obtain cross-section data, while SIMULATE3 is used to assemble and simulate all zones in 3D to obtain axial figure of merits. Multiple constraints are used to guide the optimisation search such as power peaking factor, average assembly enrichment, hot/cold k_∞, and others. Using approximately 600 CPU-hours of search, about 50 feasible bundles are found by the RL algorithm. The optimal bundle with 5 axial zones has an average burnup of 42 GWD/MTU, with variable fuel and gadolinium enrichment radially and axially.

KEYWORDS: 3D Assembly Optimisation, Deep Reinforcement Learning, BWR Neutronics, Partial length Rods, CASMO4/SIMULATE3

1. INTRODUCTION

Reducing nuclear fuel cost is one way to improve the sustainability and efficiency of the existing light water reactor fleet. For that purpose, fuel designers attempt to solve an optimization problem by utilizing expert judgement, nuclear design principles, and physics-based tools. The multi-objective, constrained, and high-dimensional nature of the optimisation problem have led nuclear utilities to rely extensively on expert judgement and manual tuning to decide the optimum assembly and core configurations. Despite significant research efforts on using evolutionary and/or stochastic algorithms like simulated annealing (SA) [1], genetic algorithms (GA) [2], and particle swarm optimisation [3] to solve these problems, expert knowledge still dominates the decision-making process. There are two main reasons behind this. First, evolutionary/stochastic algorithms feature inherent randomness in their search, which is expensive for assembly/core optimisation, given the computational costs associated with their simulation. Second, most of the commercial tools (e.g. ROSA) adopt simplified physics and surrogate models to accelerate the search, which would introduce additional uncertainties in the optimisation, as re-evaluation of the selected configurations is still needed with industry codes (e.g. CASMO4/SIMULATE3). After the recent advances of deep neural networks and reinforcement learning (RL) in autonomous control [4], RL started to propagate into other applications, specifically optimisation [5]. In our previous work [6], we explored the ability of RL (deep Q learning and proximal policy optimisation) to optimise a simplified 2D boiling water reactor (BWR) assembly model using CASMO4. The study focused on how to combine expert knowledge and learning capabilities of RL to leverage the optimisation search. Considering GA and SA as baselines, RL outperformed them through showing excellent search performance and ability to find more feasible patterns that match the predefined constraints (average enrichment, k_∞, number and location of gadolinium rods, etc.). This work utilizes our RL methodology and concepts presented in [6], however, with extension to 3D

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assembly models. A realistic 3D optimisation of the GE14-10x10 BWR assembly is performed using proximal policy optimisation (PPO) [7] and CASMO4/SIMULATE3 codes. AI-driven radial optimisation of UO_2 fuel enrichment and poison loading are performed using CASMO4 for different axial zones, while SIMULATE3 is used to assemble and simulate all zones according to a specified axial height for each zone.

2. METHODOLOGY

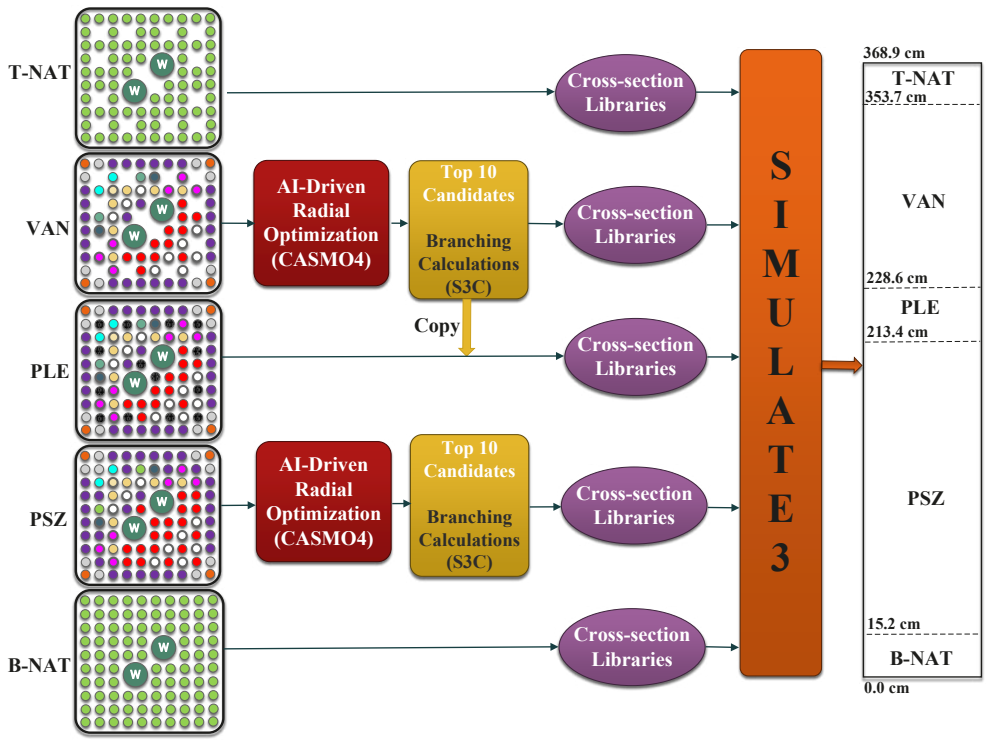

Figure 1: Five-zone 3D Optimisation of GE14-10x10 BWR assembly with RL

Figure 1 shows the methodology adopted in this work to optimise a 5-zone BWR assembly, which can be summarised as follows:

1. The bottom (B-NAT) and top (T-NAT) zones feature uniformly natural uranium enriched UO_2 fuel (0.71%), so they are not subjected to RL-driven optimisation.
2. The power shaping zone (PSZ) and the vanished zone (VAN) require radial optimisation of UO_2 fuel enrichment and gadolinium (GAD) position and concentration using CASMO4. *This step is the most challenging step in this study, since it involves solving a combinatorial optimisation problem of order of* $\sim 10^{30}$ *combinations with RL.*
3. The plenum zone (PLE) captures the plenum region of the partial-length rods. The radial fuel distribution of the PLE zone is identical to the VAN zone, except that plenum structure materials (e.g. inconel, zircaloy-2) are modeled to represent the plenum of partial-length rods (see the hashed black rods of the PLE zone in Figure 1).
4. After determining all candidates from steps 1-3, branching calculations are conducted to generate the homogenized cross-section libraries for different assembly states (rodded/unrodded, low/high void, cold/hot conditions, etc.). In particular, different void fraction values in the range 0%-70% are used to capture the boiling across the channel.

5. The selected candidates and their corresponding cross-section libraries are evaluated in 3D using SIMULATE3 according to the axial structure illustrated in the right of Figure 1. The best 3D model among all candidates that meets all pre-specified constraints is reported.

Notice that the height of all zones in Figure 1 is obtained from [8], which reported a commercial GE14-10x10 configuration. The height of PSZ zone is fixed by the height of partial-length rods. This is also true for B-NAT, PLE, and T-NAT, which also have fixed heights of about 6 in (15.24 cm). Therefore, the optimisation of zone height in SIMULATE3 is not needed for this particular 5-zone case, which simplifies the process computationally.

2.1. Fuel Types

The UO_2 fuel pellet types to choose from to optimise the zones PSZ, PLE, and VAN are as follows: 0.8%, 1.2%, 1.6%, 2.0%, 2.2%, 2.4%, 2.6%, 2.8%, 3.0%, 3.2%, 3.6%, 4.0%, 4.2%, 4.4%, 4.9%. For GAD pellets in (X%,Y%) pair, where X is U-235 enrichment and Y is Gd_2O_3 enrichment, the pellets to choose from are: (4.9%,8.0%), (4.9%,7.0%), (4.9%,6.0%), (4.4%,8.0%), (4.4%,7.0%), (4.4%,6.0%). Fuel heterogeneity between bottom and upper zones takes four different forms as follows when moving from PSZ to VAN: (1) decreasing UO_2 enrichment in UO_2 pins (e.g. 4.9% pellet in PSZ to 4.4% in VAN), (2) decreasing GAD enrichment in a GAD pellet while U-235 enrichment remains fixed (e.g. (4.9%,8.0%) pellet in PSZ to (4.9%,7.0%) in VAN), (3) decreasing both GAD and U-235 enrichment in a GAD pellet (e.g. (4.9%,8.0%) pellet in PSZ to (4.4%,7.0%) in VAN), and (4) replacing a GAD pellet with a UO_2 pellet with the same or reduced U-235 enrichment (e.g. (4.9%,8.0%) pellet in PSZ to 4.9% or 4.4% in VAN). In this work, we typically followed the axial trends in previous GE14 designs [8].

2.2. Figure of Merits

Before diving into the objective function details, we should define all figure of merits to be measured for each bundle design. In CASMO4, the following figure of merits are measured:

1. N_{GAD}: Total number of GAD pellets in the bundle regardless of the type.
2. $N_{pellets}$: Total number of "unique" pellet types of both UO_2 and GAD.
3. E: Average bundle UO_2 enrichment, which is the average of UO_2 enrichment in all fuel rods ($N_{UO_2} + N_{GAD}$). The average is calculated based on the number of fuel pellets in the zone, excluding the vanished locations. For example, for PSZ, the average is calculated based upon 92 fuel pellets, while for VAN, the average is calculated based upon 78 pellets.
4. G: Average bundle GAD enrichment, which is the mean of GAD enrichment in the bundle GAD pins (N_{GAD}). The average is calculated based upon N_{GAD}.
5. $GAD_{violate}$: a metric monitoring the placement of GAD rods in the bundle, where two GAD pellets are not allowed to be adjacent to each other in the bundle either vertically or horizontally, diagonally is allowed. $GAD_{violate}$ should be kept to zero.
6. $PPF^{x\%}$: Power peaking factor, which is the ratio of the highest fuel rod power in the bundle to the average bundle power by all rods. PPF is calculated for three different void fractions: $PPF^{0\%}$ for zero void fraction (representing bundle bottom), $PPF^{40\%}$ for 40% void fraction (representing bundle average void), and $PPF^{70\%}$ for 70% void fraction (representing bundle top).
7. $PPF^{x\%}_{rodded}$: Power peaking factor for the rodded case, which is the ratio of the highest fuel rod power in the bundle to the average bundle power by all rods, when the control rod is fully inserted. Similarly, PPF_{rodded} is calculated for the three different void fractions: 0%, 40%, and 70%, represented by $PPF^{0\%}_{rodded}$, $PPF^{40\%}_{rodded}$, and $PPF^{70\%}_{rodded}$, respectively.
8. k^{max}_{∞}: The maximum value of the neutron multiplication factor over all depletion steps.

9. $k_{\infty}^{cold,max}$: The maximum value of the neutron multiplication factor in cold conditions over all depletion steps.

10. $Burnup$: End of life burnup of the bundle, which is determined as the burnup value after the k_{∞} peak at which $k_{\infty} = 0.93$.

After radial optimisation with CASMO4, the following figure of merits in SIMULATE3 are measured for the optimal bundles: F_q (maximum pin-power peaking factor, which includes axial intranodal peaking), F_{pl} (maximum planar-averaged pin power peaking factor), F_{xy} (maximum radial pin-power peaking factor), and $F_{\Delta h}$ (maximum axially integrated pin-power peaking factor).

2.3. Core Configuration in SIMULATE3

In an attempt to capture core behaviour in this work, after completing the radial optimisation stage, the axial optimisation is conducted under a radially-reflected 2×2 micro-core simulation. This assumption is necessary here as core reload optimisation is also a complex problem and should be carefully addressed in a separate effort. As this work focuses on bundle optimisation, the small 2×2 configuration provides approximate axial power peaking factors and other metrics after moving the CASMO4 optimised zones to SIMULATE3. In SIMULATE3, two cycles are simulated to obtain more realistic power distribution in the micro-core as shown in Figure 2. For every peaking factor described above (e.g. F_q, F_{xy}), the values at all depletion steps of the second cycle are measured, see Figure 2(right), and the maximum of all values in the whole second cycle is reported for each peaking factor. We present SIMULATE3 results under *unrodded and 30% control rod insertion* during the whole cycle to obtain insights about full core performance.

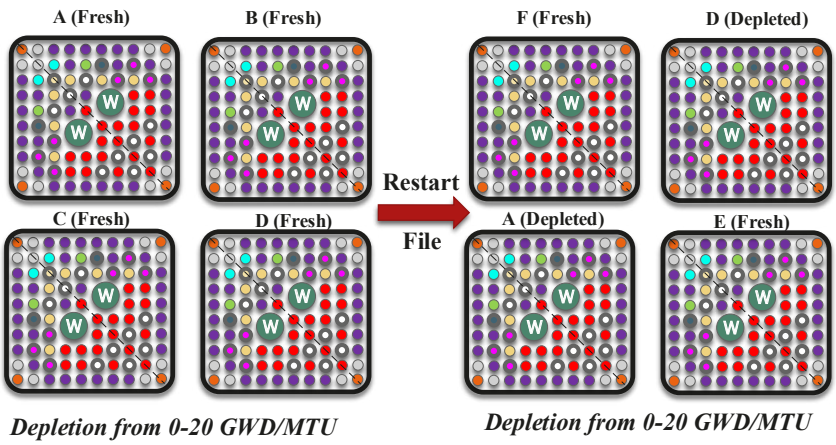

Figure 2: The 2×2 micro-core configuration in SIMULATE3

2.4. Reward Shaping

Engineering of objective functions is a difficult task, not just because there is no definition of the best function, but also because objective functions tend to be problem-dependent, and even for the same problem, decent effort is still needed to construct the objective function. Based on large number of tests, we adopt here an approach similar in concept to the ϵ-constraint approach or "divide and conquer", in which computing power is focused to meet a group of constraints, and once they are met, the optimiser moves to the next group, and so on. Once all constraints are met, the optimiser focuses on maximizing/minimizing the final objective in the set (e.g. burnup). In particular, our objective function is divided and optimised in a sequence of five steps:

Group 1 (Evaluate bundle layout): The first group optimises the following constraints: E, G, N_{GAD}, $GAD_{violate}$, and $N_{pellets}$, which all share a feature for being related to the bundle fuel configuration, and can be evaluated without a need for CASMO4 or any physics code. Therefore, this group requires wall time only and can be completed relatively fast. In this stage, the reward function takes the following form:

$$R(\vec{x}) = \max_{\vec{x}} F_1(\vec{x}) = -(V_E + V_G + V_{N_{GAD}} + GAD_{violate} + V_{pellets}), \tag{1}$$

where $\vec{x}$ is the input vector (i.e. or a flattened multi-dimensional array) representing the pellet type in every position in the bundle and V_E is the violation of matching the bundle enrichment between the desired lower/upper bounds (i.e. E_{lower}, E_{upper}), expressed as

$$V_E = \begin{cases} w(E_{lower} - E)/E_{lower}\% & E < E_{lower} \\ w(E - E_{upper})/E_{upper}\% & E > E_{upper} \\ 0 & Otherwise, \end{cases} \tag{2}$$

where $w = 100$ for V_E. Both V_G (violation of average bundle GAD enrichment) and $V_{N_{GAD}}$ (violation of number of GAD pellets in the bundle) are quantified in a similar form to V_E, except that E and the lower/upper bounds are replaced by their G and N_{GAD} correspondences plus setting $w = 100$ for V_G and $w = 1$ for $V_{N_{GAD}}$. See Table 1 for the desired range of E, G, and N_{GAD}. A special consideration for $V_{N_{GAD}}$ is given to handle a possible caveat in this constraint, where a penalty p is used to amend $V_{N_{GAD}} \leftarrow V_{N_{GAD}} + p$ based on $V_{N_{GAD}}$ value when it is calculated following the form in Eq.(2). The value of p is set to 0 if $V_{N_{GAD}} < 50$, 200 if $V_{N_{GAD}} \in [50, 100)$, 500 if $V_{N_{GAD}} \in [100, 200]$, and 1000 if $V_{N_{GAD}} > 200$.

The fourth term in Eq.(1) is expressed by $GAD_{violate} = 10 * N_{violate}$, where $N_{violate}$ is the number of GAD pins that neighbor each other vertically or horizontally. For the last term in Eq.(1), if the number of the unique pellets in the whole bundle $N_{pellets} \leq 20$, then $V_{pellets} = 0$, otherwise, $V_{pellets} = 1000$. The reader can notice two important observations about Eq.(1): (1) the minus sign is used to convert violation to reward, since in RL, the logic is built to "maximize" reward, (2) if all 5 rules in Eq.(1) are satisfied, then the maximum possible value in this group is zero. *Accordingly, if Group 1 is satisfied, i.e. $R(\vec{x}) = 0$, the pattern is recorded as **"candidate"** pattern, and it is moved to Group 2, otherwise, the pattern is discarded.*

Group 2 (Evaluate $PPF^{40\%}$): The PPF in general has been the most difficult constraint to match among all. Therefore, we first evaluate the unrodded PPF for 40% void fraction in steady state (no depletion) conditions. The reason is if the PPF violates the constraint early in cycle, then there is no need to continue analyzing this pattern after depletion. The objective function $R(\vec{x})$ for this group is switched to:

$$R(\vec{x}) = \max_{\vec{x}} F_2(\vec{x}) = 10\frac{1}{PPF^{40\%}}, \text{ if } PPF^{40\%} > PPF^{40\%,max}, \tag{3}$$

where $PPF^{40\%,max}$ is the upper limit allowed for PPF at 40% void fraction, which is indicated in Table 1 for each zone. As this group requires steady-state simulation only, each CASMO4 simulation takes on the order of 1s to complete. *If Group 2 is satisfied, i.e. $PPF^{40\%} \leq PPF^{40\%,max}$, move the bundle pattern to Group 3, otherwise, the pattern is discarded.*

Group 3 (Evaluate all PPFs): all PPFs for all void fraction values in rodded and unrodded conditions are evaluated, namely, $PPF^{0\%}$, $PPF^{40\%}$, $PPF^{70\%}$, $PPF^{0\%}_{rodded}$, $PPF^{40\%}_{rodded}$, $PPF^{70\%}_{rodded}$, still under steady state (no depletion) conditions to accelerate this group. If any of the PPF values in all zones **violates** its corresponding constraint (see Table 1), the reward is determined as

$$R(\vec{x}) = \max_{\vec{x}} F_3(\vec{x}) = 20\frac{1}{max(PPF^{0\%}, PPF^{40\%}, PPF^{70\%})} + 5\frac{1}{max(PPF^{0\%}_{rodded}, PPF^{40\%}_{rodded}, PPF^{70\%}_{rodded})}, \tag{4}$$

otherwise, this pattern is qualified to move to the next group. Notice that in Eqs.(3)-(4), we have used the reciprocal to convert our goal of minimizing PPF to reward maximization, and more weight is given to PPF than PPF_{rodded} since it is more important. While still a steady-state simulation, the multiple voids and rodded states increase run time to approximately 10s. *If Group 3 is satisfied (i.e. all PPF and PPF_{rodded} constraints are met), move the bundle*

pattern to Group 4, otherwise, discard the pattern.

Group 4 (Evaluate k_∞^{max}/$k_\infty^{cold,max}$***)***: now the pattern is depleted to a maximum of 50 GWD/MTU. First, the PPF constraints are verified again after depletion to ensure no violation occurs. Afterward, the maximum k_∞^{max} and $k_\infty^{cold,max}$ are estimated. If any of the k_∞^{max}/$k_\infty^{cold,max}$ values in all zones violates their corresponding constraint (see Table 1), the reward of this group can be estimated as

$$R(\vec{x}) = \max_{\vec{x}} F_4(\vec{x}) = 10\frac{1}{\overline{k_\infty^{max}}} + 10\frac{1}{\overline{k_\infty^{cold,max}}}, \tag{5}$$

otherwise, this pattern is qualified to move to the next group. Here $\overline{k_\infty^{max}}$ and $\overline{k_\infty^{cold,max}}$ are the average of k_∞^{max} and $k_\infty^{cold,max}$ in all active zones involved in the optimisation, respectively. Here, notice again that we use the reciprocal to convert our goal of minimizing $k_\infty^{max}/k_\infty^{cold,max}$ to reward maximization. Adding depletion increases the simulation cost of CASMO4 to approximately 2 min. *If Group 4 is satisfied (i.e. all* k_∞^{max} *and* $k_\infty^{cold,max}$ *constraints are met), move the bundle pattern to Group 5, otherwise, the pattern is discarded.*

Group 5 (Evaluate $Burnup$***)***: Finally, based on the same depletion calculation performed in Group 4, if all design constraints are satisfied, its burnup is determined based on $k_\infty = 0.93$ in the depletion curve (An authors' choice, which can be increased or decreased depending on the application). The final form of the reward function becomes

$$R(\vec{x}) = \max_{\vec{x}} F_5(\vec{x}) = Burnup - 10\max(PPF^{0\%}, PPF^{40\%}, PPF^{70\%}) \tag{6}$$

Therefore, the final objective in this work is to search for bundle design with maximum burnup and minimum PPF, while all safety and design constraints are met. We call the patterns that satisfy Groups 1-4 as **"feasible"** patterns, **while the pattern with maximum** $R(\vec{x}) = F_5(\vec{x})$ **is reported in this work**. By observing Table 1, the reader can conclude that we have a total of 23 constraints to meet. These constraints impose significant challenge to solve this optimisation problem as all constraints have to be respected simultaneously.

2.5. Proximal Policy Optimisation

Proximal policy optimisation (PPO) [7] is selected as the RL algorithm in this work for multiple reasons related to stability, easier implementation and parallelization, less hyperparameter sensitivity, and its popularity in the RL research. PPO trains a policy that directly maps *states* (the bundle fuel distribution in current time step) to appropriate *actions* to take (how the pellet types should be selected in the next time step).

PPO hyperparameters according to different resources should be tuned for optimal performance, see [7] for more details. Based on grid search tuning and our prior experience with PPO in the stable-baselines package for RL [9], the optimal PPO hyperparameters used in this work are only listed here, while the reader is referred to this reference for hyperparameter description [9]. The optimal hyperparameters are: `policy` =MlpPolicy with 2 hidden layers, fully connected with 64 nodes per layer, `gamma` =0.99, `n_steps` =380, `n_env` =25 parallel processors, `ent_coef` =0.01, `learning_rate` =2.5×10^{-4}, `vf_coef` =0.5, `lam` =0.9, `nminibatches` =4, `noptepochs` =15, `cliprange` =0.2, and the remaining parameters were set to default. Considering PPO with the hyperparameters mentioned above in stable-baselines, the RL training scenario proceeds as follows:

1. Each episode consists of 500 time steps. In the first step, a random state (s_1) is generated, which is nothing but a random sample drawn from the action/fuel space.
2. The PPO agent decides the first action (a_1), which highlights the fuel pellets in the active zones.
3. All zones are evaluated using CASMO4, and the reward of this pattern (r_1) is evaluated according to Eqs.(1)-(6).
4. The sampled action is assigned to the next state (i.e. $s_2 \leftarrow a_1$), then the next state (s_2) and the reward (r_1) are returned back to the PPO agent for another step.

5. Steps 1-4 are repeated until 500 time steps are completed (i.e. 500 bundles are evaluated). At this time, the episode is terminated, and a new episode starts.

6. Step 5 is repeated until all episodes (or total number of time steps) are completed.

Two major notes should be highlighted about PPO. First, after the first episode, we start the next episodes with the best state of highest reward from previous episodes, as this change demonstrated training improvement. Second, once 25 episodes (i.e. 25000 time steps) are completed, their best rewards are grouped together into 1 epoch, as this grouping allows us to draw more accurate conclusions about the agent performance using the epoch statistics (e.g. mean, standard deviation, max, min), compared to monitoring every time step or every episode.

Table 1: List of the desired constraints considered for the bundle radial optimisation of the 2 active zones PSZ and VAN

Variable	Value
E (1 constraint)*	$[3.95\%, 4.05\%]$
G (1 constraint)*	$[7.2\%, 7.6\%]$
N_{GAD} (2 constraints)	[16,18], [13,16]
$GAD_{violate}$ (2 constraint)	0, 0
$N_{pellets}$ (1 constraint)	≤ 20
$PPF^{0\%}$ (2 constraints)	$\leq 1.45, \leq 1.45$
$PPF^{40\%}$ (2 constraints)	$\leq 1.4, \leq 1.4$
$PPF^{70\%}$ (2 constraints)	$\leq 1.4, \leq 1.4$
$PPF^{0\%}_{rodded}$ (2 constraints)	$\leq 2.0, \leq 2.0$
$PPF^{40\%}_{rodded}$ (2 constraints)	$\leq 1.9, \leq 1.9$
$PPF^{70\%}_{rodded}$ (2 constraints)	$\leq 1.9, \leq 1.9$
k_∞^{max} (2 constraints)	$\leq 1.1, \leq 1.15$
$k_\infty^{cold,max}$ (2 constraints)	$\leq 1.18, \leq 1.2$

* Constraint represents the average of both zones together.

3. RESULTS

Figure 3 demonstrates detailed RL/PPO convergence plots for the reward as well as other relevant figure of merits. Figure 3(a)-(d) show various statistics for each figure of merit from reward to burnup. Each epoch consists of 25000 reward/objective calculations or 3D assembly patterns. Over 80 epochs, the total number of evaluations is 2.0 millions. Therefore, for every 25000 evaluations, the mean, standard deviation, max, and min are estimated and plotted. Figure 3(e)-(f) show only cumulative sum of candidate and feasible patterns, as cumulative sum is more informative for these two figure of merits. Typical learning curves can be observed in Figure 3. The reward starts from a deeply negative value due to violating Group 1 layout constraints in Eq.(1). After matching Group 1 constraints (e.g. E, G), the number of candidate patterns starts to rise in Figure 3(e). Afterward, the reward keeps building up until reaching the ultimate objective in Eq.(6), where the number of feasible patterns in Figure 3(f) also increases. As can be observed, the search converges to regions where all constraints are met. For example, the max PPF of all zones and all void fraction values in Figure 3(c) falls below 1.45, the max of all predefined PPF thresholds in Table 1. In addition, the average UO_2 enrichment E converges into the desired range [3.95%-4.05%] in Figure 3(b). This is also applicable to the rest of the figure of merits.

For the best/optimal design, the summary of the CASMO4 figure of merits is given in Table 2. The following figure of merits are recorded for the 3D bundle in SIMULATE3 for **unrodded case**: $F_{xy} = 1.50$, $F_q = 2.69$, $F_{pl} = 2.67$, $F_{\Delta_h} = 1.34$, while the following values are for the 30% rodded case: $F_{xy} = 1.69$, $F_q = 2.72$, $F_{pl} = 2.7$, $F_{\Delta_h} = 1.46$. The radial/axial fuel distribution is shown in Figure 4. The results are presented for all zones in each design (active

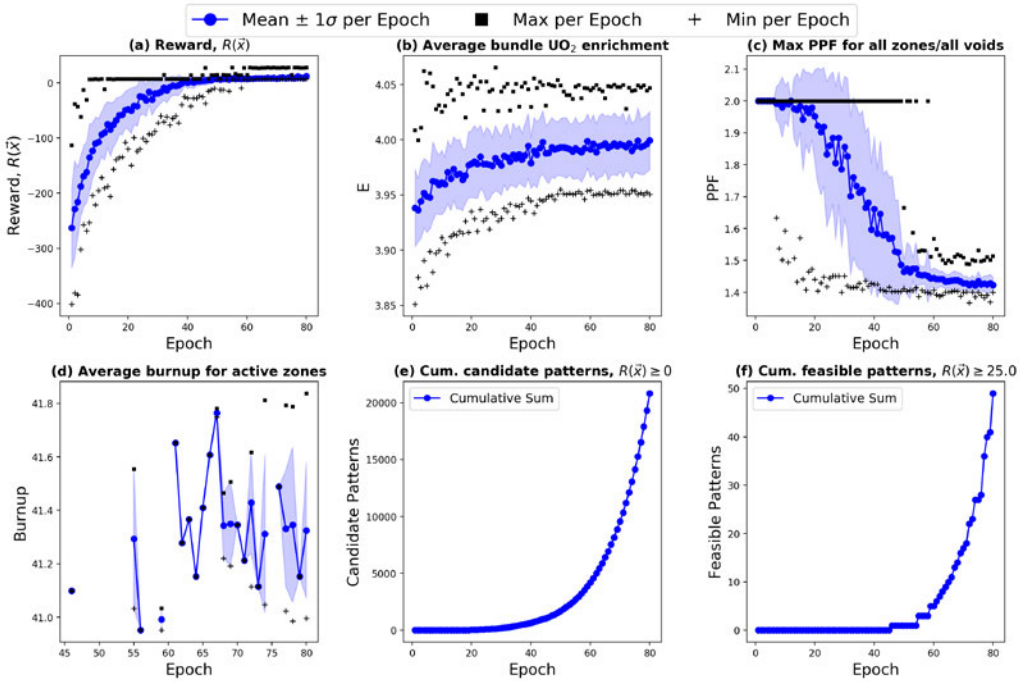

Figure 3: Learning curves for radial optimisation of the bundle active zones. One epoch involves 25000 time steps or evaluated bundles

and inactive). The burnup is not reported for B-NAT/T-NAT/PLE zones due to their small heights. By comparing the results in Table 2, we can see that this optimised design satisfies all constraints specified in Table 1. The optimal design features a total of **17 unique pellets** with radial and axial fuel heterogeneity, see Figure 4. For example, the reader can notice in Figure 4 that GAD concentration has decreased from 8% in PSZ zone to 7% or 6% in VAN zone.

In any optimisation problem, adding constraints makes the search more difficult, and for this work, having about 23 constraints to satisfy is the main challenge. However, some of the problem constraints originate from expert knowledge such as restricting N_{GAD} into a certain range (e.g. 16-18), spacing out GAD pins in each zone, and favoring low enrichment at bundle boundary, which all help the optimiser to reach the feasible region (i.e. high burnup and low PPF). Given the combinatorial nature of this problem and its exponential dependency, O(m^n), reasonable choices of the fuel pins (m) by the analyst can play a dramatic role in the optimisation progress and performance, given that n value here is fixed to 51 positions. Logically, searching in a space with lower dimensionality will increase the likelihood of finding competitive feasible patterns than in extremely high-dimensional spaces with many choices.

Obtaining varieties of feasible patterns can be very helpful once these bundles are moved to full core analysis, which is not investigated in this work. As BWR cores have 5-15 unique assembly types, it is expected that the core designer would use a mixture of these feasible patterns to obtain optimal core performance. Overall, we did not highlight the burnup of the SIMULATE3 core model, as assembly burnup in the 2×2 micro-core configuration is not informative, and core burnup of 600-700 assemblies needs to be estimated. In this work, we resolved some of the previous gaps in reactor optimisation literature [10][11] by capturing the lattice physics details in upper zones during optimisation, which give the designer more optimal performance after moving the bundle into the core. Considering three different void fraction values (0%, 40%, 70%), rodded, and cold conditions give more realistic representation of the fission rate at bundle bottom, middle, and upper zones compared to not using them. Additionally, our work considers wide range of pin types for both UO_2 and GAD pins, allowing more flexible radial/axial optimisation, as typically needed to achieve robust optimisation.

Table 2: CASMO4 figure of merits of the optimal bundle design in Figure 4

Item	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5
Zone ID	B-NAT	PSZ[a]	PLE	VAN[a]	T-NAT
Zone height (cm)	15.2	198.2	15.2	125.1	15.2
N_{GAD}	0	16	14	14	0
E^b	0.71%	4.07%	3.97%	3.97%	0.71%
G^b	0	8.00%	6.85%	6.85%	0
$GAD_{violate}$	0	0	0	0	0
$PPF^{0\%}$	1.19	1.37	1.32	1.34	1.13
$PPF^{40\%}$	1.20	1.37	1.35	1.34	1.15
$PPF^{70\%}$	1.21	1.39	1.37	1.36	1.15
$PPF^{0\%}_{rodded}$	1.68	1.80	1.74	1.74	1.62
$PPF^{40\%}_{rodded}$	1.72	1.69	1.65	1.65	1.65
$PPF^{70\%}_{rodded}$	1.75	1.61	1.60	1.59	1.67
k_∞^{max}	0.89659	1.09189	1.12942	1.14215	0.88393
$k_\infty^{cold,max}$	0.85933	1.15898	1.16654	1.16782	0.81102
$\boldsymbol{Burnup}$	**-**	**42.7**	**-**	**40.9**	**-**

[a] Active zone
[b] The bundle active zone average is within the constraints in Table 1

4. SUMMARY

Application of optimisation algorithms to nuclear assembly design has been a topic of interest over the years to improve fuel efficiency, reduce costs, while meeting safety limits. In this work, we present an application of deep reinforcement learning (RL) for large-scale optimisation of BWR bundles. A 3D bundle optimisation is conducted considering both radial and axial heterogeneity in the fuel loading as well as many design constraints that are considered in the real-world BWR bundle design process. Using approximately 600 CPU-hours of search, about 50 feasible bundles in 3D form are found by the RL algorithm. The optimal bundle with 5 axial zones has an average burnup of 42 GWD/MTU, with variable fuel and gadolinium enrichment radially and axially. As this work was limited to a 2×2 micro-core simulation of the bundle in SIMULATE3, future work will focus on combining the 3D bundle optimisation with full core, in which the feasible patterns found in this study can be tested in a multi-cycle full core simulation to analyze their overall safety and economic performance.

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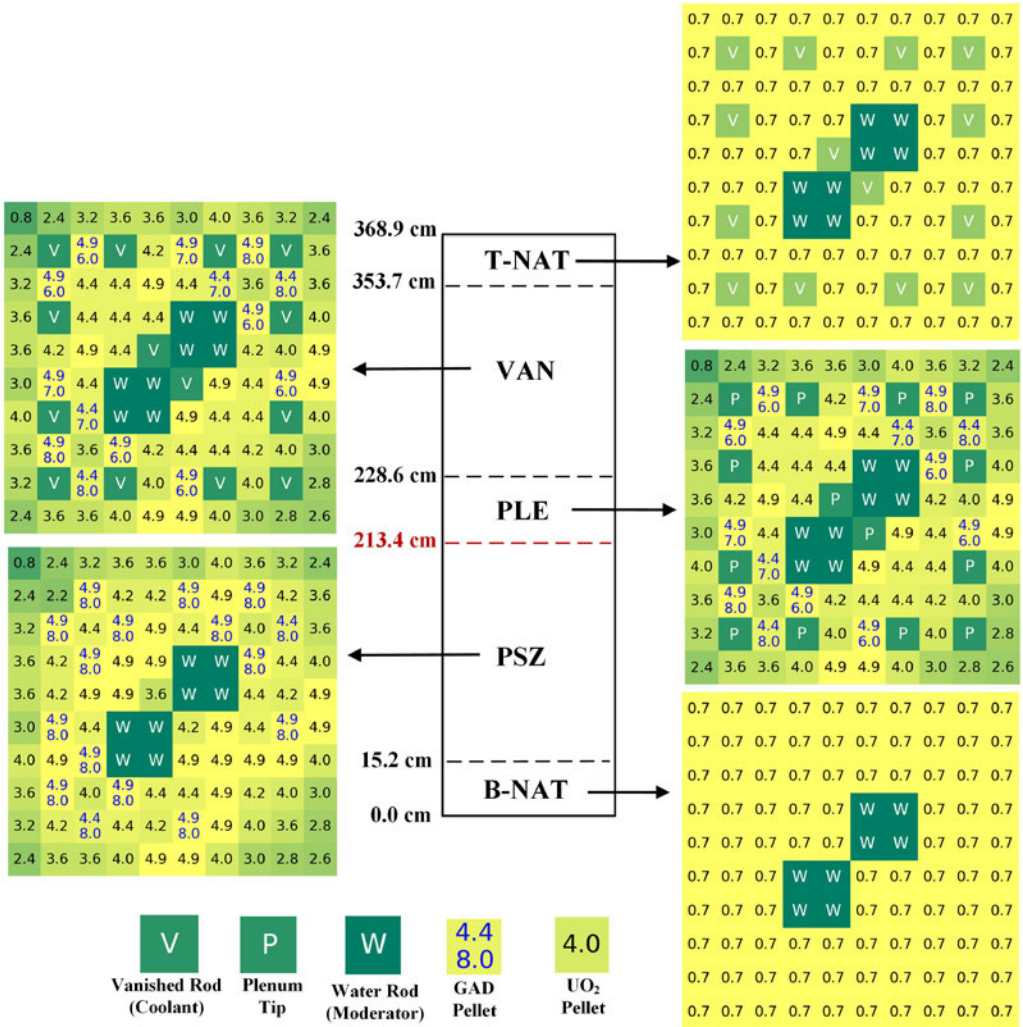

Figure 4: Radial and axial fuel distributions of the best optimised bundle

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